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PENIS MEASUREMENT SYSTEM AND METHOD

Abstract

A penis measurement system and method, wherein the system includes a user device and a penis measurement device. An external soft casing encases a main shell of the penis measurement device. When at least one strap of the external soft casing is strapped on a penis, the external soft casing fixes the main shell to be on top of the penis, allowing a detection airbag protruding from the main shell to press down onto the penis with a down-pressing pressure. The detection airbag deforms and presses onto a sensor chip in the main shell. A processing unit receives a pressure sensor signal from the sensor chip and a gyroscope signal from a gyroscope in the main shell, and accordingly outputs a hardness value and a gyroscope value to a user device through a communication unit. The system quantifies a hardness of the penis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the priority benefit of TW application serial No.113107578 filed on Mar. 1, 2024, the entirety of which is hereby incorporated by reference herein and made a part of the specification.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a penis measurement system and method, capable of more accurately detecting and recording male physiological data during sexual activities.

2. Description of the Related Art

[0003] For the purpose of avoiding any misunderstandings, the present application refers to males and females as traditionally defined biological males and biological females.

[0004] In the technical field of gender physiological studies, a hardness of penis determines whether a sexual intercourse with a female can be successful. The hardness of penis, however, currently cannot be unbiasedly, scientifically, quantitatively gauged, but only qualitatively rated. For this reason, professionals in the said technical field need an unbiased apparatus to obtain the hardness of penis for the purpose of scientifically evaluating the successfulness of the penis to have sexual intercourse.

[0005] In the technical field of gauging a hardness of a piece of rubber, traditionally the rubber is down-pressed with a force, and then upon release, a force of the rubber bouncing back is gauged and measured. This hardness measurement of the piece of rubber is often conducted by a Shore durometer with a C-shaped protruding tip. The protruding tip of the Shore durometer is internally connected to a fixed spring, thus by down-pressing the protruding tip into an object with a fixed depth and measuring how much the protruding tip is being pushed back by the object, the hardness of the object is gauged and measured. In other words, a height of the object pushing back the protruding tip is measured by the Shore durometer. More particularly, by manually or using machine to down-press the protruding tip 90 degrees into a surface of the object, the surface of the object would bounce back the protruding tip and exert pressure of the fixed spring. The distance of the object pushing back the protruding tip is positively correlated to how much force is exerted on the fixed spring. As a result, the harder the object is, the shallower the protruding tip is able to be pressed into the surface of the object, and the more distance the object pushes back the protruding tip, thus adding more pressure on the fixed spring of the Shore durometer and causing the Shore durometer to display a greater measurement value. Vice versa, the softer the object is, the deeper the protruding tip is able to be pressed into the surface of the object, and the less distance the object pushes back the protruding tip, thus putting lesser pressure on the fixed spring of the Shore durometer and causing the Shore durometer to display a lesser measurement value. In short, the greater the pressure exerted by the surface of the object bouncing back, the greater measurement value is measured by the Shore durometer, thus signifying the object having a greater measured hardness.

[0006] In the technical field of measuring the hardness of penis, currently the following problems remain to be solved.

[0007] Problem 1: when attempting to measure the hardness of penis, a detection of the male nocturnal penile tumescence (NPT) cannot be effectively conducted. According to known

physiological studies, males tend to have NPT naturally during rapid-eye-movement (REM) sleeps, and the hardness of the penis as well as the counts of erections during NPT are viewed as standard unbiased data usable for evaluating a male physiological status. In a traditional practice, this hardness measurement is conducted by attaching a stamp paper to a root of the penis at night. In the next morning, if the stamp paper is torn by the penis, the penis is hastily concluded to have enough hardness, when enlarging for having an erection during NPT, to overcome a resistance force imposed by the stamp paper to the penis. However, this traditional practice only measures a degree of enlargement of the penis in a single instance, thus unable to provide statistical data for erection counts. The traditional practice also cannot accurately and quantitatively measure the degree of enlargement of the penis as well as the hardness of the penis.

[0008] A current alternative to measure the penis is by using Rigiscan®. However, although Rigiscan® theoretically is able to measure erection counts, in practice, a fixing contraption of the Rigiscan® often slips away from the penis during a man's sleep. Furthermore, the Rigiscan® is too large in size, thus when the man turns during sleep, the Rigiscan® often prevents the man from sleeping comfortably, and as a result of bad sleep, the results measured by the Rigiscan® may not most accurately represent the data gathered for an erection during NPT.

[0009] Due to the above reasons, the Rigiscan® still leaves desirable improvements for one to more effectively conduct measurements for the erection of the penis during NPT.

[0010] Problem 2: currently the hardness of a penis cannot be scientifically measured. A traditional clinical way of measuring the hardness of a penis often uses erectile hardness score (EHS). The EHS is first published in *The Journal of Sexual Medicine*, a famous medical journal, under “Validation of the Erection Hardness Score” as a research report by scholar Mulhall in 2007. Subsequently, Pfizer, a famous biopharmaceutical company, adopts the results published by Mulhall in the research report into four degrees of erectile hardness meters and created four corresponding penis models to qualitatively represent the four different degrees of erectile hardness. As a result, when a man self-describes the hardness of his penis in retrospect, the four penis models are provided to aid the man in qualitatively identifying the hardness of his penis. In other words, the man may self-identify his penis to have the same hardness of one of the four options. However, such a patient-reported outcome (PRO) is highly susceptible to being biased, and the biological sampling method is also not well-regulated. An individual may be hazy about his own penis hardness, resulting in extremely wide margin of error, especially when compared to other individual's own qualitative measurements. For the above reasons, currently the hardness of a penis cannot be scientifically measured, and the measurement of the hardness of a penis needs further improvements.

[0011] For reference, according to the research paper: Miranda, E., Hallan, B., Nascimento, B., Paul, G., Carneiro, F., De Bessa Junior, J., & Miranda, A. (2023). (121) “Validation of a Digital Rigidometer for the Evaluation of Erection Hardness During In-office Rigidity Assessment in Patients with Erectile Dysfunction,” *The Journal of Sexual Medicine*, 20, when a penis cannot overcome a first degree (equivalent of 0.51 kilograms, or roughly about 5 Newtons) of down-pressing obstacle force or a second degree (equivalent of 0.75 kilograms, or roughly about 7.35 Newtons) of down-pressing obstacle force, the penis would bend and be unable to overcome the obstacle force imposed by a vagina, thus the penis would not be able to successfully have a sexual intercourse with the vagina. However, when the penis is able to overcome at least third degree (equivalent of 1.2 kilograms, or roughly about 11.76 Newtons) of down-pressing obstacle force, the penis would be able to remain straight and hard for overcoming the obstacle force imposed by the vagina and have a successful sexual intercourse. The down-pressing obstacle force is stress exerted by surfaces of the vagina towards a tip of the penis facing the vagina.

[0012] Problem 3: when using the Rigiscan® to measure the penis, the Rigiscan® is only able to measure up to roughly 60% of the hardness of the enlarged penis. According to the research paper: Allen, R. P., Smolev, J. K., Engel, R. M., & Brendler, C. B. (1993). “Comparison of RigiScan and

formal nocturnal penile tumescence testing in the evaluation of erectile rigidity.” *The Journal of Urology*, 149 (5), 1265-1268, the Rigiscan® apparatus includes a belt bonding a root part of the penis and a corona of glans of the penis. By investigating a change of tightness of the belt, the Rigiscan® is used to measure the hardness of the penis. However, according to research responses from the research paper: Guay, A. T., & Heatley, G. J. (1994). “Re: Comparison of RigiScan and formal nocturnal penile tumescence testing in the evaluation of erectile rigidity.” *The Journal of Urology*, 152 (1), 171-171, a discovery is made by the researcher that minor abnormality of the erected penis cannot be accurately measured when the penis is enlarged to exceed 60% of the hardness measurement (measurement of the tightness of the belt) from the apparatus. For this reason, the Rigiscan® cannot be considered as an accurate apparatus for measuring the hardness of the penis.

[0013] Problem 4: currently no apparatus is capable of measuring the hardness of the penis during a sexual intercourse. According to the research paper: Timm, G. (1999). “Axial penile buckling forces vs Rigiscan (TM) radial rigidity as a function of intracavernosal pressure: why Rigiscan does not predict functional erections in individual patients-Editorial comment.” *INTERNATIONAL JOURNAL OF IMPOTENCE RESEARCH*, 11 (6), 338-339, although an axial hardness of the penis is discussed by the article, no practical methodology is introduced for measuring the hardness of the penis during a sexual intercourse. For this reason, the research paper only presents theoretical discussions, and does not propose a practical solution of measuring the hardness of the penis during a sexual intercourse.

SUMMARY OF THE INVENTION

[0014] To overcome the aforementioned problems, the present invention provides a penis measurement system and method, capable of unbiasedly, scientifically, and quantitatively measuring a hardness data of the penis in a sexual intercourse and presenting a solution to the aforementioned problems.

[0015] The penis measurement system of the present invention includes:

[0016] a penis measurement device, configured to be fixed on a penis of a user, and including:

[0017] a main shell, including a sensor opening; [0018] a detection airbag, positioned inside of the main shell, and protruding outwards from the sensor opening; [0019] a sensor chip, mounted inside of the main shell, and contacting the detection airbag; [0020] a communication unit, mounted inside of the main shell; [0021] a processing unit, mounted inside of the main shell, and electrically connected to the sensor chip and the communication unit; [0022] an external soft casing, encasing the main shell, and including at least one strap configured to be fixed on the penis; [0023] wherein when the at least one strap is strapped on the penis, the external soft casing fixes the main shell to be on top of a root part of a penis, the detection airbag contacts the penis through the external soft casing, and the detection airbag, from top of the root part of the penis, presses down onto the penis with a down-pressing pressure; [0024] wherein when the detection airbag receives a bounce-back pressure corresponding to the down-pressing pressure from the penis, the detection airbag deforms and presses onto the sensor chip, thus allowing the sensor chip to generate a pressure sensor signal corresponding to a stress exerted onto the sensor chip by the detection airbag; the sensor ship sends the pressure sensor signal to the processing unit, and the processing unit calculates and outputs a hardness value according to the pressure sensor signal;

[0025] a user device, communicatively connected to the communication unit of the penis measurement device; wherein the user device receives the hardness value outputted by the processing unit via the communication unit of the penis measurement device.

[0026] Since the external soft casing fixes the main shell to be on top of the root part of the penis when the at least one strap of the penis measurement device is strapped on the penis, when the user is having a sexual intercourse, the penis measurement device on the penis would not hinder the penis from moving in back and forth motions. Simultaneously, the detection airbag that is contacting the penis from top of the penis is able to receive and detect the bounce-back pressure

corresponding to the down-pressing pressure.

[0027] To generalize, the harder the penis is, the less the penis would change its shape when receiving the down-pressing pressure, hence the penis would produce the greater bounce-back pressure corresponding to the down-pressing pressure. As the detection airbag receives the greater bounce-back pressure, the detection airbag would deform more to squeeze more onto the sensor chip, thus exerting more stress onto the sensor chip. As the sensor chip receives the greater stress, the sensor chip outputs the greater pressure sensor signal to the processing unit, thus allowing the processor unit to calculate the greater hardness value. In short, when the detection airbag is squeezed by the penis, the higher the detection airbag pressure is raised by the squeeze, the greater the detection airbag would deform and squeeze the sensor chip, hence representing having the greater hardness of the penis.

[0028] Vice versa, the softer the penis is, the greater the penis would change its shape when receiving the down-pressing pressure, hence the penis would produce the lesser bounce-back pressure corresponding to the down-pressing pressure. As the detection airbag receives the lesser bounce-back pressure, the detection airbag would deform less to squeeze less onto the sensor chip, thus putting lesser stress onto the sensor chip. As the sensor chip receives the lesser stress, the sensor chip outputs the lesser pressure sensor signal to the processing unit, thus allowing the processor unit to calculate the lesser hardness value.

[0029] As a result, the penis measurement device is able to overcome the aforementioned problems 1 to 4 for assisting the user to unbiasedly, scientifically, and quantitatively measure the hardness value of the penis when having a sexual intercourse. Furthermore, as the user device is able to receive the hardness value, more subsequent applications and calculations relating to the hardness value may be provided to the user through the user device.

[0030] The penis measurement device also includes a gyroscope. The gyroscope is configured for monitoring a degree of back and forth moving motion of the user's buttocks when the user is having a sexual intercourse. The gyroscope is mounted inside of the main shell, and the gyroscope is electrically connected to the processing unit. The gyroscope generates a gyroscope signal according to movement changes of the penis measurement device and sends the gyroscope signal to the processing unit. The processing unit calculates a gyroscope value according to the gyroscope signal, and the processing unit controls the communication unit to send the gyroscope value to the user device. In an embodiment, the user device is a smart phone.

[0031] A penis measurement method of the present invention is executed by the user device, and the penis measurement method includes the following steps:

[0032] recording a data receiving time for receiving a hardness value and a gyroscope value outputted from a penis measurement device;

[0033] uploading the hardness value, the gyroscope value, and the data receiving time to a cloud server;

[0034] continuously receiving the hardness value, the gyroscope value, and the data receiving time, and constructing a data changing graph according to the hardness value, the gyroscope value, and the data receiving time; and

[0035] displaying the data changing graph through a display.

[0036] As such, the data generated from the user having the sexual intercourse is able to be simultaneously uploaded and stored in the cloud server. The user may subsequently access the stored data in the cloud server. The user may also conveniently look at the data changing graph displayed on the display for gaining insights towards the data changes relating to the sexual intercourse over a period of time. This allows the user to unbiasedly analyze a history of the user's physiological changes during the period of time when the user is having the sexual intercourse.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] FIG. **1** is a block diagram of an embodiment of a penis measurement system of the present invention.

[0038] FIG. **2** is an exterior perspective view of a penis measurement device of the penis measurement system of the present invention.

[0039] FIG. **3** is an exploded perspective view of the penis measurement device of the penis measurement system of the present invention.

[0040] FIG. **4** is another exploded perspective view of the penis measurement device of the penis measurement system of the present invention.

[0041] FIG. **5** is a side-perspective view of the penis measurement device of the penis measurement system of the present invention.

[0042] FIG. **6** is a cross-sectional perspective view of the penis measurement device of the penis measurement system of the present invention.

[0043] FIG. **7** is a perspective view of using the penis measurement device of the penis measurement system of the present invention.

[0044] FIG. **8** is another perspective view of using the penis measurement device of the penis measurement system of the present invention.

[0045] FIG. **9** is a flow chart of a penis measurement method of the present invention.

[0046] FIG. **10** is another flow chart of a penis measurement method of the present invention.

[0047] FIG. **11** is another flow chart of a penis measurement method of the present invention.

[0048] FIG. **12** is a perspective view of a user device of the penis measurement system of the present invention displaying a first option page.

[0049] FIG. **13** is a perspective view of the user device of the penis measurement system of the present invention displaying a second option page.

[0050] FIG. **14** is a perspective view of a user device of the penis measurement system of the present invention displaying a calibration to zero page.

[0051] FIG. **15** is a perspective view of a user device of the penis measurement system of the present invention displaying a real-time data page.

[0052] FIG. **16** is a perspective view of a user device of the penis measurement system of the present invention displaying a data result page.

[0053] FIG. **17** is a perspective view of a user device of the penis measurement system of the present invention displaying a historical records page.

DETAILED DESCRIPTION OF THE INVENTION

[0054] The present invention provides a penis measurement system and method, capable of assisting a user to unbiasedly, scientifically, and quantitatively measure a hardness of penis when having sexual activities, such as when having a sexual intercourse. As such, the present invention is able to resolve the following problems: a problem that patient-reported outcome (PRO) of self-assessed penis hardness is highly susceptible to being biased and cannot be scientifically adopted; a problem that when using the Rigiscan® to measure the penis, the Rigiscan® is only able to measure up to roughly 60% of the hardness of the enlarged penis; and a problem of currently having no capable apparatus for measuring the hardness of the penis during a sexual intercourse.

[0055] With reference to FIG. **1**, the present invention provides a penis measurement system. The penis measurement system includes a penis measurement device **100** and a user device **200**. The user device **200** is communicatively connected to the penis measurement device **100**. The penis measurement device **100** is configured to be fixed on the user's penis, thus assisting to effectively measure physiological data related to the penis. Furthermore, regardless of what conditions the user is in, for example, regardless of the user is sleeping soundly and having nocturnal penile tumescence (NPT), or the user is fantasizing and masturbating, or the user is having a sexual intercourse with his partner, the present invention would be able to unbiasedly measure the

physiological data related to the penis.

[0056] With reference to FIGS. 2, 3, and 6, the penis measurement device **100** includes a main shell **110**, an external soft casing **120**, and a detection airbag **130**. The main shell **110** includes a sensor opening **111**. The detection airbag **130** is positioned inside of the main shell **110**, and the detection airbag **130** protrudes outwards from the sensor opening **111** away from the main shell **110**. The external soft casing **120** encases the main shell **110**, and the external soft casing **120** includes at least one strap configured to be fixed on the penis. The penis measurement device **100** also includes a processing unit **10**, a sensor chip **20**, and a communication unit **30** that are mounted inside of the main shell **110**. The sensor chip **20** contacts the detection airbag **130**, the communication unit **30** is communicatively connected to the user device **200**, and the processing unit **10** is respectively electrically connected to the sensor chip **20** and the communication unit **30**.

[0057] In an embodiment, the at least one strap of the external soft casing **120** includes a first strap **121** and a second strap **122**, as well as a first tightness regulator **123** corresponding to the first strap **121** and a second tightness regulator **124** corresponding to the second strap **122**. The first strap **121** and the second strap **122** are both hollow polymer closed-loop straps. The first strap **121** and the second strap **122** are respectively connected to two opposite sides of the external soft casing **120** that correspond to a detection position **125** of the detection airbag **130**. In addition, the first strap **121** is configured to pass through the first tightness regulator **123**, and the first tightness regulator **123** may slide along the first strap **121** for regulating a tightness of the first strap **121**. The second strap **122** is configured to pass through the second tightness regulator **124**, and the second tightness regulator **124** may slide along the second strap **122** for regulating a tightness of the second strap **122**. In other embodiments, the at least one strap of the external soft casing **120** may have different numbers of straps and strap configurations.

[0058] With reference to FIGS. 5, 7, and 8, when the first strap **121** or the second strap **122** is tied to a penis **300**, for example, when the penis measurement device **100** of the present invention is strapped to be on top of the penis **300** by the first strap **121**, the second strap **122**, the first tightness regulator **123** corresponding to the first strap **121**, and the second tightness regulator **124** corresponding to the second strap **122**, the main shell **110** is fixed to be on top of a root part **1** of the penis **300**. Simultaneously, the detection airbag **130** contacts the penis **300** through the external soft casing **120**, and the detection airbag **130**, from top of the root part **1** of the penis **300**, presses down onto the penis **300** with a down-pressing pressure

[0059] In an embodiment, the meaning of the detection airbag **130** contacting the penis **300** through the external soft casing **120**, is that the detection airbag **130** extends outside of an opening of the external soft casing **120** to directly contact the penis **300**. As such, the present invention is able to most simply and directly conduct measurements on the penis **300**. In another embodiment, the meaning of the detection airbag **130** contacts the penis **300** through the external soft casing **120**, is that the detection airbag **130** indirectly contacts the penis **300** through the external soft casing **120**. As such, when conducting measurements on the penis **300**, the external soft casing **120** is able to keep the sensor chip **20** inside of the main shell **110** dry from moisture and to keep the detection airbag **130** clean. In the present embodiment, the external soft casing **120** is completely waterproof.

[0060] This allows the external soft casing **120** to be easily cleaned with water solutions and stay sanitized.

[0061] More particularly, the first strap **121** and the second strap **122** respectively extend towards two different directions from the two opposite sides of the detection position **125** on the external soft casing **120**. The first strap **121** and the second strap **122**, extending towards two different directions, form a difference angle θ in between. In an embodiment, the difference angle θ is 30 degrees.

[0062] By having the difference angle θ optimized for the first strap **121** and the second strap **122** to tighten around the penis **300**, the present invention is able to make sure that when the penis **300** is moving back and forth with the user's buttocks, i.e., when the user is having a sexual intercourse,

the first strap **121** and the second strap **122** can be firmly fixed on the penis **300**, ensuring that the penis measurement device **100** is free from being loose or falling from the root part **1** of the penis **300**. By having the difference angle θ optimized for the first strap **121** and the second strap **122** to tighten around the penis **300**, the present invention also avoids becoming an obstacle hindering the penis **300** from moving back and forth in motions. To ensure the penis measurement device **100** is securely fastened on the penis, when the first strap **121** and the second strap **122** are fastened on the penis **300**, as shown in FIG. 7 and FIG. 8, the first strap **121** and the second strap **122** are respectively fastened by the first tightness regulator **123** and the second tightness regulator **124** below a bottom part **2** of the penis and above a top part **3** of a testicle/testicles. When the penis **300** is moving in back and forth motions, since the first tightness regulator **123** and the second tightness regulator **124** are fixed on the top part **3** of the testicles, the testicles would not be hurt, hindered, or disturbed by the first tightness regulator **123** and the second tightness regulator **124**. In comparison to an apparatus of Rigiscan®, the external soft casing **120** is able to structurally allow the detection airbag **130** to conduct measurement on the penis **300** without hindering the sexual intercourse process. Furthermore, the external soft casing **120** also prevents the penis measurement device **100** from falling during a man's sleep. As the external soft casing **120** is free from hurting the testicles, the man would be able to sleep comfortably. **8**

[0063] About the detection airbag **130** conducting measurements on the penis **300**, the detection airbag **130** contacting the penis on the root part **1** of the penis **300** is able to receive a bounce-back pressure corresponding to the down-pressing pressure from the penis **300**. When the detection airbag **130** receives the bounce-back pressure corresponding to the down-pressing pressure, the detection airbag **130** deforms and squeezes (presses) onto the sensor chip **20**, thus allowing the sensor chip **20** to generate a pressure sensor signal corresponding to a stress exerted onto the sensor chip **20** by the detection airbag **130**. The sensor chip **20** then sends the pressure sensor signal to the processing unit **10**, and the processing unit **10** calculates and outputs a hardness value according to the pressure sensor signal. The hardness value is the quantified quantity representing the hardness of the user's penis.

[0064] The harder the penis **300** is, the less the penis would deform when receiving the down-pressing pressure, hence the penis **300** would produce the greater the bounce-back pressure corresponding to the down-pressing pressure. As the detection airbag **130** receives the greater the bounce-back pressure, the detection airbag **130** would deform more to squeeze more onto the sensor chip **20**, thus putting more stress onto the sensor chip **20**. As the sensor chip **20** receives the greater stress, the sensor chip **20** outputs the greater pressure sensor signal to the processing unit **10**, thus allowing the processor unit **10** to calculate the greater hardness value.

[0065] Vice versa, the softer the penis **300** is, the greater the penis **300** would deform when receiving the down-pressing pressure, hence the penis **300** would produce the lesser bounce-back pressure corresponding to the down-pressing pressure. As the detection airbag **130** receives the lesser bounce-back pressure, the detection airbag **130** would deform less to squeeze less onto the sensor chip **20**, thus putting lesser stress onto the sensor chip **20**. As the sensor chip **20** receives the lesser stress, the sensor chip **20** outputs the lesser pressure sensor signal to the processing unit **10**, thus allowing the processor unit **10** to calculate the lesser hardness value.

[0066] As the apparatus of Rigiscan® is not completely accurate in measuring the hardness and the enlargement of the penis, for example, as described in prior art that the Rigiscan® is only able to measure up to roughly 60% of the hardness of the enlarged penis, the present invention uses a completely different measuring principle. The present invention utilizes a hardness measuring principle used by a Shore durometer, thus measuring the hardness of the penis with better accuracy, in other words, more accurately gauging the bounce-back ability of the penis after the penis endures the down-pressing pressure.

[0067] The hardness value of the present invention is originated from the pressure sensor signal generated and outputted by the sensor chip **20**. In the present embodiment, the sensor chip **20** is a

piece of micro electro mechanical systems (MEMS) configured for detecting pressure. For example, the sensor chip **20** may be a pressure sensor of model LPS22HBTR manufactured by Mouser Electronics®. The said pressure sensor uses a strain gauge as a pressure sensitive component, and the stress sensed by the strain gauge is the pressure forced onto the sensor chip **20** by the detection airbag **130**. Although the hardness value of the present invention originated from the pressure changes sensed by the sensor chip **20**, when the sensor chip **20** is structurally configured with the rest of components in the penis measurement device **100**, the hardness value calculated by the present invention retains the physical meaning of quantifying the hardness of the penis. The hardness value calculated by the present invention and value gauged by Shore hardness are positively correlated.

[0068] Furthermore, in an embodiment, the penis measurement device **100** further includes an eccentric rotating mass (ERM) motor **40**, a gyroscope **50**, a charging port **60**, a battery unit **70**, a buttons unit **80**, and a memory unit **90**. The ERM motor **40**, the gyroscope **50**, the charging port **60**, the battery unit **70**, the buttons unit **80**, and the memory unit **90** are respectively electrically connected to the processing unit **10**, and the charging port **60** is further electrically connected to the battery unit **70**. The ERM motor **40**, the gyroscope **50**, and the battery unit **70** are mounted inside of the main shell **110**, and the charging port **60** and the buttons unit **80** are mounted on the main shell **110**. With reference to FIG. **6**, in the present invention, the penis measurement device **100** further includes a circuit board **101**. The processing unit **10**, the communication unit **30**, the gyroscope **50**, and the memory unit **90** are all mounted on the circuit board **101**. The processing unit **10** is a processor, and the memory unit **90** is a memory.

[0069] When the ERM motor **40** is started by the processing unit **10**, the ERM motor **40** produces vibrations, thus vibrating the penis measurement device **100**.

[0070] The gyroscope **50** generates a gyroscope signal according to movement changes of the penis measurement device **100** and sends the gyroscope signal to the processing unit **10**. The processing unit **10** calculates a gyroscope value according to the gyroscope signal. The meaning of the gyroscope value will be explained in later parts of the description. In general, regardless of the user wearing the penis measurement device **100** wearing any kinds of conscious or unconscious movements relating to the penis **300**, such as having the sexual intercourse or the NPT, the gyroscope **50** would be able to record movement and directional changes of the penis measurement device **100** into the gyroscope signal, and send the gyroscope signal to the processing unit **10** for the processing unit **10** to calculate the gyroscope value that is invaluable for analyzing physiological status of the user. By having the gyroscope **50**, the back and forth motions of the buttocks of the user may be additionally monitored during the sexual intercourse.

[0071] After calculating the hardness value and the gyroscope value, the processing unit **10** may control the communication unit **30** for communicatively connecting to the user device **200**, thus having the communication unit **30** sending the hardness value and the gyroscope value to the user device **200**. In an embodiment, the communication unit **30** of the penis measurement device **100** uses Bluetooth® to communicatively connect to the user device **200**. As the user device **200** receives the hardness value and the gyroscope value from the penis measurement device **100**, the user device **200** may conduct further calculations and data processing procedures using its own settings, thus providing the user with additional information regarding the penis of the user. The user May examine the data collected by the penis measurement device **100** through the user device **200**. In an embodiment, when the processing unit **10** controls the ERM motor **40** to produce vibrations, since the vibration emanating from the ERM motor **40** does not significantly displace the penis measurement device **100**, the gyroscope **50** would still be able to count the back and forth motions of the penis measurement device **100** unaffected.

[0072] The battery unit **70** stores electrical charges for the penis measurement device **100**. The battery unit **70** may be rechargeable batteries, and the charging port **60** is configured to electrically connect an external power source for charging the battery unit **70**.

[0073] With further reference to FIG. 4, the external soft casing **120** includes a hole **126**, and the hole **126** corresponds to a position of the charging port **60**. When the external soft casing **120** encases the main shell **110**, the charging port **60** on the main shell **110** is configured to position in the hole **126**, thus allowing the charging port **60** to be exposed for charging.

[0074] In an embodiment, the user device **200** is a computer system, and therefore, the user device **200** includes a system unit, a memory, a display, a keyboard, a mouse, and a module responsible for communicating with the communication unit **30** of the penis measurement device **100**. The user device **200** and the communication unit **30** of the penis measurement device **100** may be connected through physical ports and hard-wires, such as connected through Ethernet ports or universal serial bus (USB) ports.

[0075] In another embodiment, the user device **200** is a smart personal device, such as a smart phone, a tablet computer, or a smart wearable device. In the present embodiment, the user device **200** is a smart phone, thus the user device **200** includes a touch screen, a memory, and a wireless communication module. The user device **200** uses the wireless communication module to communicate with the communication unit **30** of the penis measurement device **100**. In the embodiment, the touch screen is the display of the user device **200**. The memory of the user device **200** stores a plurality of mode options, and the user may select one of the mode options to control the penis measurement device **100**. The user device **200** may also connect to a cloud server through the communication unit **30**, allowing the user device **200** to upload and store the physiological data measured from the penis measurement device **100**.

[0076] With reference to FIG. 9, the present invention also provides a penis measurement method. The penis measurement method is executed by a user device of the penis measurement system. The penis measurement method includes the following steps:

[0077] step **S10**: recording a data receiving time for receiving a hardness value and a gyroscope value outputted from a penis measurement device;

[0078] step **S20**: uploading the hardness value, the gyroscope value, and the data receiving time to a cloud server;

[0079] step **S30**: continuously receiving the hardness value, the gyroscope value, and the data receiving time, and constructing a data changing graph according to the hardness value, the gyroscope value, and the data receiving time; and

[0080] step **S40**: displaying the data changing graph through a display.

[0081] In an embodiment, the data receiving time is recorded according to a system time of the user device **200** at the moment the user device **200** receives the hardness value and the gyroscope value.

[0082] With reference to FIG. 10, in an embodiment, before step **S10**, the penis measurement method further includes the following steps:

[0083] step **S1**: wirelessly connecting to the communication unit of the penis measurement device;

[0084] step **S2**: selecting a working mode of the penis measurement device from the plurality of mode options;

[0085] step **S3**: calibrating zero for a sensor chip of the penis measurement device with a return-to-zero procedure;

[0086] step **S4**: determining whether finishing return-to-zero procedure; when the return-to-zero procedure is unfinished, executing step **S3**; and step **S5**: when the return-to-zero procedure is finished, starting to accept receiving the hardness value and the gyroscope value outputted from the penis measurement device, and further executing step **S10**.

[0087] With reference to FIG. 11, in an embodiment, the step **S10** further includes the following sub-steps:

[0088] step **S11**: when receiving the hardness value and the gyroscope value, starting recording the data receiving time and starting counting an activity time;

[0089] step **S12**: analyzing changes of the gyroscope value, according to a physiological model data, for generating and updating an erection counter and a flap counter; and

[0090] step **S13**: determining whether receiving a stop measurement command; when without receiving the stop measurement command, executing step **S12**; when receiving the stop measurement command, executing step **S20**.

[0091] When executing step **S20**, the present invention may also execute the following step: uploading the erection counter and the flap counter accumulated by the user device to the cloud server, and displaying the erection counter, the flap counter, and the activity time on the display of the user device.

[0092] In the present embodiment, the aforementioned user device **200** is a smart phone, and the smart phone uses an application (APP) to execute the penis measurement method of the present invention for controlling the penis measurement device **100**. With reference to the aforementioned descriptions, the following examples are demonstrated in tandem with perspective views of screenshots of software interfaces from the APP that is executed by the smart phone.

[0093] With reference to FIG. **12**, the user is able to look at the information presented by the APP displayed on the touch screen of the user device **200** and use the touch screen to select a working mode from a plurality of mode options in the APP. In an embodiment, on a first option page **210** of the APP that is displayed by the touch screen of the user device **200**, multiple options are presented to the user for choosing an initiation option of the penis measurement device **100**, thus configuring the penis measurement device **100** to work in tandem with the user. Through various pages of the APP, displayed by the touch screen of the user device **200**, mainly the user device **200** is able to provide the user with a data result of the measurement on the penis. The data result can then be subsequently used by the user or professionals for evaluation and analysis. For example, the mode options provided to the user through the APP for selecting the initiation option includes a participation with my own hand option **211**, a participation with my partner's hand(s) option **212**, a partake in sexual intercourse option **213**, and a partake in NPT detection option **214**. The initiation option selected by the user would influence parameters of how the user device **200** controls the penis measurement device **100** to conduct measurements of the penis.

[0094] In an embodiment, when the partake in NPT detection option **214** is selected as the initiation option, and once the penis measurement device **100** communicates to the user device **200** that the penis measurement device **100** is ready for conducting measurements, the user device **200** would disconnect its Bluetooth connection from the penis measurement device **100**. After being communicatively disconnected from the user device **200**, the penis measurement device **100** would store the measurements, i.e. the hardness value and the gyroscope value, of the penis over a period of time in the memory unit **90**. When the user wakes up the next morning, and when the user uses the user device **200**, such as the smart phone, to communicatively re-connect with the penis measurement device **100**, the penis measurement device **100** would send the hardness value and the gyroscope value accumulated over the period of time in the memory unit **90** to the user device **200** via Bluetooth connection.

[0095] Subsequently, the smart phone would then upload the hardness value and the gyroscope value for the period of time to a database of the cloud server. As such, physiological data of monitoring the penis over a night is recorded and uploaded to the cloud server online. More particularly, the period of time monitoring the hardness value and the gyroscope value, refers to a time when the penis measurement device **100** starts taking data to a time when the user wakes up and switches off the penis measurement device **100**. When the penis measurement device **100** receives a switch off command and prepares to shut down, the penis measurement device **100** stops recording the hardness value and the gyroscope value. When the penis measurement device **100** is subsequently switched on, the penis measurement device **100** then proceeds to send the hardness value and the gyroscope value gathered over the period of time to the user device **200**.

[0096] With reference to FIG. **13**, after the participation with my own hand option **211** is selected for the initiation option, the APP of the user device **200** displays a second option page **220** through the touch screen. The mode options provided to the user for selecting a partaking option includes a

partake in relaxation option **221**, a partake in fantasizing option **222**, and a partaking watching porn option **223**. The partaking option selected by the user provides the professionals, such as sex therapist, with invaluable evidence for analyzing data and subsequently guiding the user to improve sexual activities and physiological status.

[0097] With reference to FIG. **14**, after the partaking watching porn option **223** is selected for the partaking option, the APP of the user device **200** displays a calibration to zero page **230** through the touch screen. The calibration to zero page **230** includes a zeroing calibration option **231** and a preparation complete option **232**. Simultaneously, the calibration to zero page **230** also includes a wearing instruction information **233** for guiding the user to wear the penis measurement device **100**. When the zeroing calibration option **231** is selected, the user device **200** thus acknowledges that the user has put on the penis measurement device **100** by following the wearing instruction information **233**, and the user device **200** proceeds to control the penis measurement device **100** for calibration with the aforementioned return-to-zero procedure.

[0098] When the return-to-zero procedure is still on-going, the preparation complete option **232** on the calibration to zero page **230** is thus still unavailable for the user to select. When the return-to-zero procedure is completed, only then does the user device **200** make the preparation complete option **232** on the calibration to zero page **230** available for the user to select. For example, in the example shown in FIG. **14**, the preparation complete option **232** covered in dots symbolizes that the preparation complete option **232** is still unavailable for selection. When the preparation complete option **232** is available for selection, then the preparation complete option **232** would be spotless as shown for the zeroing calibration option **231** on the calibration to zero page **230**. When the preparation complete option **232** is selected, the user device **200** thus acknowledges that the user has completed all preparations and is ready for measurements on the penis, and the user device **200** proceeds to control the penis measurement device **100** for conducting the measurements on the hardness of the penis and gather other physiological data from the user. The other physiological data, for example, as previously disclosed, may be obtaining the erection counter and the flap counter by analyzing changes of the gyroscope value according to the physiological model data stored in the user device **200**.

[0099] With reference to FIG. **15**, after the preparation complete option **232** is selected, the penis measurement device **100** starts delivering the hardness value and the gyroscope value to the user device **200** in real-time. The user device **200** thus starts obtaining real-time measurements of the hardness value and the gyroscope value taken by the penis measurement device **100**, and the APP of the user device **200** displays a real-time data page **240** through the touch screen. The user device **200** executes the aforementioned steps of the penis measurement method for displaying an activity time **241**, a hardness value **242**, a hardness data changing graph **243** corresponding to changes of the hardness value **242** over time, and a flapping speed changing graph **244** corresponding to changes of the gyroscope value over time on the real-time data page **240**. In the example shown in FIG. **15**, the hardness value **242** is the hardness value measured by the sensor chip **20** over each second. In other words, the hardness value **242** has a denominator as one second (1/second, or 1/s), and the hardness value **242** is measured as the hardness value/s. A vertical axis of the hardness data changing graph **243** represents the hardness value **242**, and a horizontal axis of the hardness data changing graph **243** represents chronological progression of the activity time **241**. A vertical axis of the flapping speed changing graph **244** records a flapping speed. The flapping speed, for example, is measured as the gyroscope value per second, thus may be represented as (the gyroscope value)/s. A horizontal axis of the flapping speed changing graph **244** also represents chronological progression of the activity time **241**. As the activity time **241** and the hardness value **242** are updated in real-time, the hardness data changing graph **243** and the flapping speed changing graph **244** are also continuously being updated. The vertical axis and horizontal axis of both the hardness data changing graph **243** and the flapping speed changing graph **244** would be continuously adjusted for optimizing scales presented to the user. The continuous update and adjustments of the

hardness data changing graph **243** and the flapping speed changing graph **244** present immediate physiological feedbacks to the user while the user is having sexual activities with the penis. In general, towards different degrees and ways of back and forth body movements, or towards different degrees and ways of flapping, the present invention would measure and analyze the hardness value and the gyroscope value differently. The present invention does not limit the ways of measuring and analyzing the hardness value and the gyroscope value of the present invention. The emphasis of the present invention, is that through parameters configured by the APP, the sensor chip **20** and the gyroscope **50** of the penis measurement device **100** may be programmed to measure the hardness value and the gyroscope value differently. In an embodiment, the hardness value is X/hPa (wherein X is the numeric measured value, and hPa is a pressure unit of a hundred Pascal). In an embodiment, the gyroscope value may be measured in a numeric count of back and forth motion over a minute or over a second, or in other words, (the numeric count of a flap)/min or (the numeric count of a flap)/s. The numeric count of a flap is the flap counter.

[0100] In an embodiment, the real-time data page **240** also displays a stop option **245**. When the stop option **245** is selected, the user device **200** immediately generates the stop measurement command, thus the user device **200** stops the penis measurement device **100** from measuring the penis according to the stop measurement command.

[0101] With reference to FIG. **16**, once stopping measuring, the APP of the user device **200** displays a data result page **250** through the touch screen. The data result page **250** retains the most recent update of the activity time **241**, the hardness data changing graph **243**, and the flapping speed changing graph **244**.

[0102] The user device **200** further executes the following steps: recording a total time as the activity time **241** last updated by the user device **200**, and displaying the total time;

[0103] obtaining a highest value of the vertical value of the hardness data changing graph **243** across all times, setting the highest value as a highest hardness value, and displaying the highest hardness value;

[0104] calculating a total peak number by counting all the peaks of the vertical values of flapping speed changing graph **244** that are greater than a threshold, setting the total peak number as a numeric count of a flap, and displaying the numeric count of the flap, or in other words, displaying the flap counter.

[0105] For example, corresponding to the above steps, a highest hardness value **251** and a numeric count **252** of a flap are displayed on the data result page **250**.

[0106] With reference to FIG. **17**, the touch screen of the user device **200** may, at any given time, be switched to display a historical records page **260** of the APP. The historical records page **260** displays historical records of the user's physiological data stored in the database of the cloud server. Each record of the user's physiological data on the historical records page **260** includes a record time **261**, a total time **262**, and a mode information **263**. The record time **261** is obtained from a central time synchronized by the smart phone and the cloud server, and the record time **261** is also the aforementioned data receiving time recorded by the penis measurement device **100** for receiving hardness value and gyroscope value. The total time **262** is the aforementioned total time for recording the said record of the user's physiological data, in other words, the total time **262** is the activity time **241** last updated for the said record of the user's physiological data. The mode information **263** corresponds to the initiation option chosen by the user through the user device **200** for configuring how the penis measurement device **100** should conduct measurements of the penis. Each record of the user's physiological data on the historical records page **260** may be selected by the user through interacting with the touch screen for revealing more detailed recorded physiological information.

[0107] Overall, the user of the present invention is able to obtain unbiased, scientific, and quantitative physiological information of the penis under various application settings through wearing the penis measurement device **100**. The physiological information of the penis gathered by

the present invention, for example, may include the hardness value of the penis when the user is having a sexual intercourse. Furthermore, the present invention provides the user an easy convenience of accessing the historical records of the user's physiological data measurements through the user device **200** accessing the database of the cloud server. For example, the user may access to read about the highest hardness value **251** of the penis that is analyzed for a training practice of the penis or a sexual intercourse of the penis. These invaluable data are able to assist the user oneself, doctors, or professionals to more effectively, more accurately, and more scientifically measure the hardness of the user's penis, thus providing invaluable insights over the user's physiological status, and providing assistive information for the doctors or the professionals to decide on further treatments or training courses related to the penis for the user.

[0108] For reference, please consider the following research papers:

[0109] Research paper 1: Ku, J., Song, Y., Kim, M., Lee, N., & Park, Y. (2001). "Is there a role of radial rigidity in the evaluation of erectile dysfunction?" *International journal of impotence research*, 13 (4), 200-204; and

[0110] Research paper 2: Rohrer, G. E., Premo, H., & Lentz, A. C. (2022). "Current Techniques for the Objective Measures of Erectile Hardness." *Sexual Medicine Reviews*, 10 (4), 648-659.

[0111] According to the research papers 1 and 2, the research papers 1 and 2 support that the hardness of the penis along a radial direction and a resistive force along the radial direction are significantly correlated, and thus the hardness value of the penis that is measured by the present invention is indeed positively correlated to how much resistance force is measured by the present invention. The measurement obtained by the present invention is able to help further investigating the physiological research of: exactly how hard the penis needs to be for the penis to successfully overcome the resistive force imposed by the vagina in order for the penis to successfully enter into the vagina.

Claims

1. A penis measurement system, comprising: a penis measurement device, configured to be fixed on a penis of a user, and comprising: a main shell, comprising a sensor opening; a detection airbag, positioned inside of the main shell, and protruding outwards from the sensor opening; a sensor chip, mounted inside of the main shell, and contacting the detection airbag; a communication unit, mounted inside of the main shell; a processing unit, mounted inside of the main shell, and electrically connected to the sensor chip and the communication unit; an external soft casing, encasing the main shell, and comprising at least one strap configured to be fixed on the penis; wherein when the at least one strap is strapped on the penis, the external soft casing fixes the main shell to be on top of a root part of the penis, the detection airbag contacts the penis through the external soft casing, and the detection airbag, from top of the root part of the penis, presses down onto the penis with a down-pressing pressure; wherein when the detection airbag receives a bounce-back pressure corresponding to the down-pressing pressure from the penis, the detection airbag deforms and presses onto the sensor chip, thus allowing the sensor chip to generate a pressure sensor signal corresponding to a stress exerted onto the sensor chip by the detection airbag; the sensor ship sends the pressure sensor signal to the processing unit, and the processing unit calculates and outputs a hardness value according to the pressure sensor signal; a user device, communicatively connected to the communication unit of the penis measurement device; wherein the user device receives the hardness value outputted by the processing unit via the communication unit of the penis measurement device; a gyroscope, mounted inside of the main shell, and electrically connected to the processing unit; wherein the gyroscope generates a gyroscope signal according to movement changes of the penis measurement device and sends the gyroscope signal to the processing unit; wherein the user device calculates a gyroscope value according to the gyroscope signal, and the processing unit controls the communication unit to send the gyroscope

value to the user device.

2. The penis measurement system as claimed in claim 1, wherein the at least one strap of the external soft casing comprises a first strap and a second strap; the first strap and the second strap are hollow polymer closed-loop straps; the first strap and the second strap are respectively connected to two opposite sides of the external soft casing that correspond to a detection position of the detection airbag; wherein the first strap and the second strap respectively extend towards two different directions from the two opposite sides of the detection position on the external soft casing; wherein when the first strap and the second strap are fastened on the penis, the first strap and the second strap are fastened below a bottom part of the penis and above a top part of a testicle/testicles.

3. The penis measurement system as claimed in claim 2, wherein the at least one strap of the external soft casing also comprises a first tightness regulator corresponding to the first strap and a second tightness regulator corresponding to the second strap; wherein the first strap is configured to pass through the first tightness regulator, and the first tightness regulator slides along the first strap for regulating a tightness of the first strap; wherein the second strap is configured to pass through the second tightness regulator, and the second tightness regulator slides along the second strap for regulating a tightness of the second strap.

4. The penis measurement system as claimed in claim 1, wherein the penis measurement device further comprises: a battery unit, mounted inside of the main shell, electrically connected to the processing unit, and storing electrical charges; a charging port, mounted on the main shell, and electrically connected to the processing unit and the battery unit; wherein the external soft casing comprises a hole, and the hole corresponds to a position of the charging port; when the external soft casing encases the main shell, the charging port on the main shell is configured to position in the hole; an eccentric rotating mass (ERM) motor, mounted inside of the main shell, and electrically connected to the processing unit; wherein when the ERM motor is started by the processing unit, the ERM motor produces vibrations.

5. The penis measurement system as claimed in claim 1, wherein the sensor chip is a piece of micro electro mechanical systems (MEMS) configured for detecting pressure.

6. The penis measurement system as claimed in claim 1, wherein the user device is a smart phone, and the smart phone is wirelessly connected to the communication unit of the penis measurement device; wherein the smart phone executes an application (APP) with a software interface for controlling the penis measurement device.

7. The penis measurement system as claimed in claim 6, wherein the penis measurement device comprises a memory unit, and the memory unit is electrically connected to the processing unit; wherein when a partake in nocturnal penile tumescence (NPT) detection option is selected through the software interface of the APP, and once the penis measurement device communicates to the user device that the penis measurement device is ready for conducting measurements, the user device disconnects from the penis measurement device, and the penis measurement device stores the hardness value and the gyroscope value over a period of time in the memory unit; wherein when the user device communicatively re-connects with the penis measurement device, the penis measurement device sends the hardness value and the gyroscope value accumulated over the period of time in the memory unit to the user device wirelessly.

8. The penis measurement system as claimed in claim 6, wherein the software interface of the APP simultaneously presents a zeroing calibration option, a preparation complete option, and an instruction information for guiding the user to wear the penis measurement device; wherein when the zeroing calibration option is selected, the user device proceeds to calibrate the penis measurement device; wherein after the user device calibrates the penis measurement device, when the preparation complete option is selected, the user device proceeds to control the penis measurement device for taking measurements on the penis.

9. The penis measurement system as claimed in claim 6, wherein the penis measurement device

delivers the hardness value and the gyroscope value to the user device in real-time; wherein the user device displays a graphical representation of changes of the hardness value in real-time through the APP.

10. A penis measurement method, executed by the user device as claimed in claim 1, and comprising the following steps: recording a data receiving time for receiving a hardness value and a gyroscope value outputted from a penis measurement device; uploading the hardness value, the gyroscope value, and the data receiving time to a cloud server; continuously receiving the hardness value, the gyroscope value, and the data receiving time, and constructing a data changing graph according to the hardness value, the gyroscope value, and the data receiving time; and displaying the data changing graph through a display.

11. The penis measurement method as claimed in claim 10, wherein the step of recording the data receiving time for receiving the hardness value and the gyroscope value comprises the following sub-steps: when receiving the hardness value and the gyroscope value, starting recording the data receiving time and starting counting an activity time; analyzing changes of the gyroscope value, according to a physiological model data stored in the user device, for generating and updating an erection counter and a flap counter; and when receiving a stop measurement command, stopping the penis measurement device from measuring the penis.

12. The penis measurement method as claimed in claim 11, wherein after stopping measuring the penis, further comprising the following steps: uploading the erection counter and the flap counter accumulated by the user device to the cloud server, and displaying the erection counter, the flap counter, and the activity time on the display of the user device.

13. The penis measurement method as claimed in claim 11, further comprising the following steps: recording a total time as the activity time last updated by the user device, and displaying the total time; obtaining a highest value of the hardness value across all times, setting the highest value as a highest hardness value, and displaying the highest hardness value.

14. The penis measurement method as claimed in claim 10, wherein before recording the data receiving time, comprising the following steps: wirelessly connecting to the communication unit of the penis measurement device, and selecting a working mode of the penis measurement device from the plurality of mode options; calibrating zero for a sensor chip of the penis measurement device with a return-to-zero procedure, and when the return-to-zero procedure is finished, starting to accept receiving the hardness value and the gyroscope value outputted from the penis measurement device.
